Cell Biology

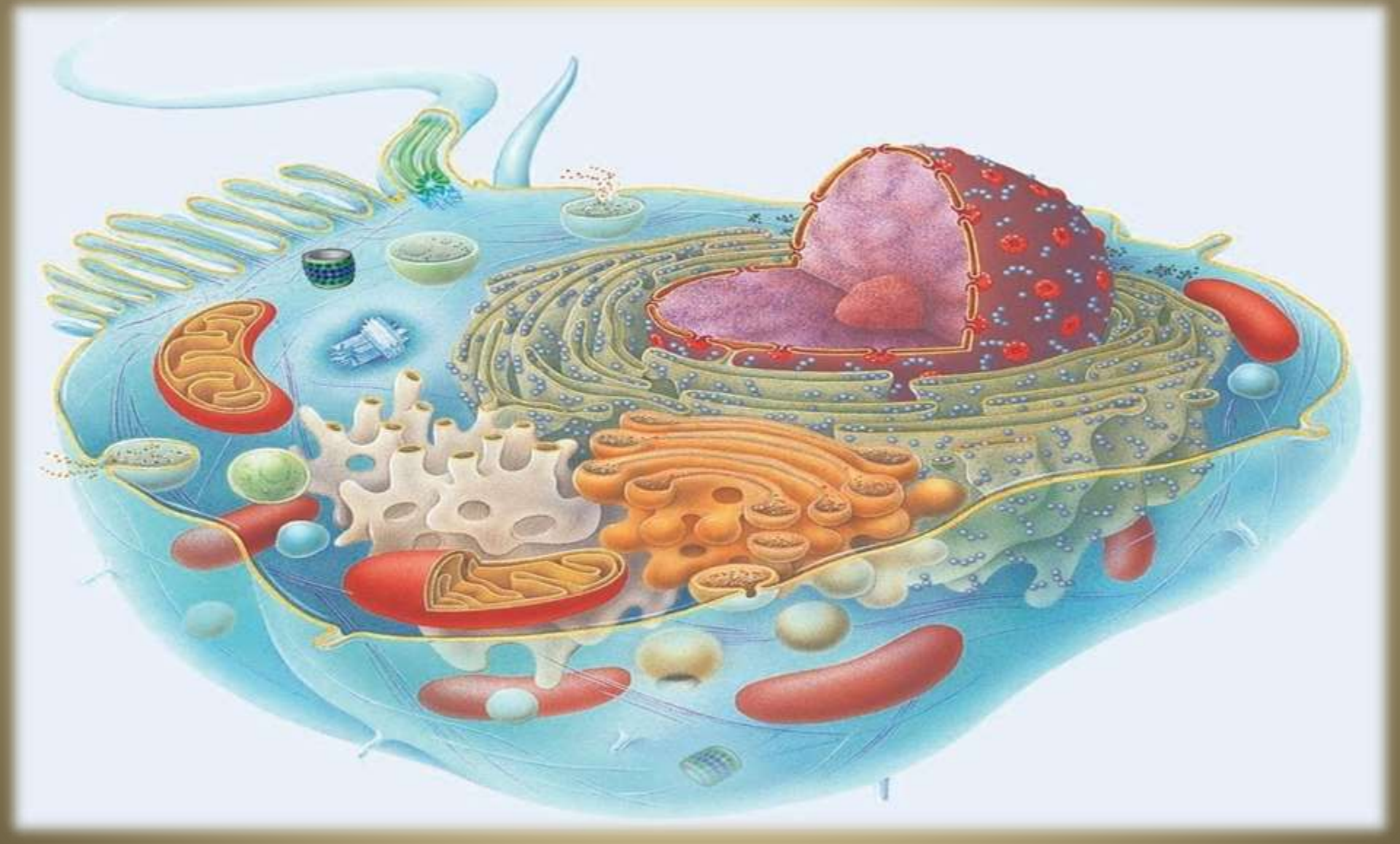
1st year BDS

Cell

- **Cell - Latin "small room"**
- **Functional and Structural unit of all living organisms.**

Cell was first discovered by

Robert Hooke in 1665.



Modern Cell theory-

- Cells make up all living matter.
- All cells arise from other cells.
- Chemical reactions of cell, anabolism and catabolism take place inside the cell.

Thus, cell is the fundamental unit of life.

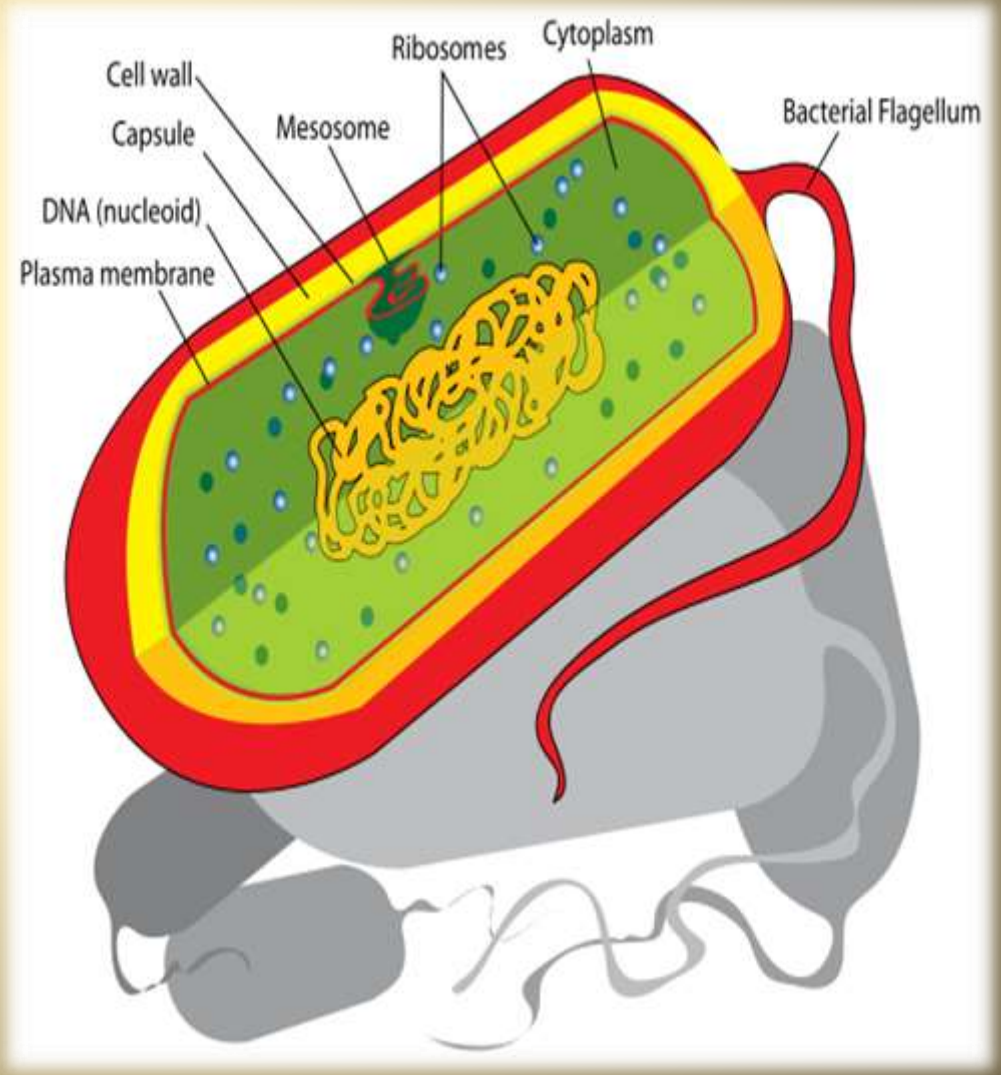
Types of cells

- Prokaryotic - lack a nucleus or membrane-bound structures
- Ex- Bacteria.

- Eukaryotic - have a nucleus and membrane-bound organelles .
- Ex- Fungi, Plants, Animals.

Prokaryotic cells

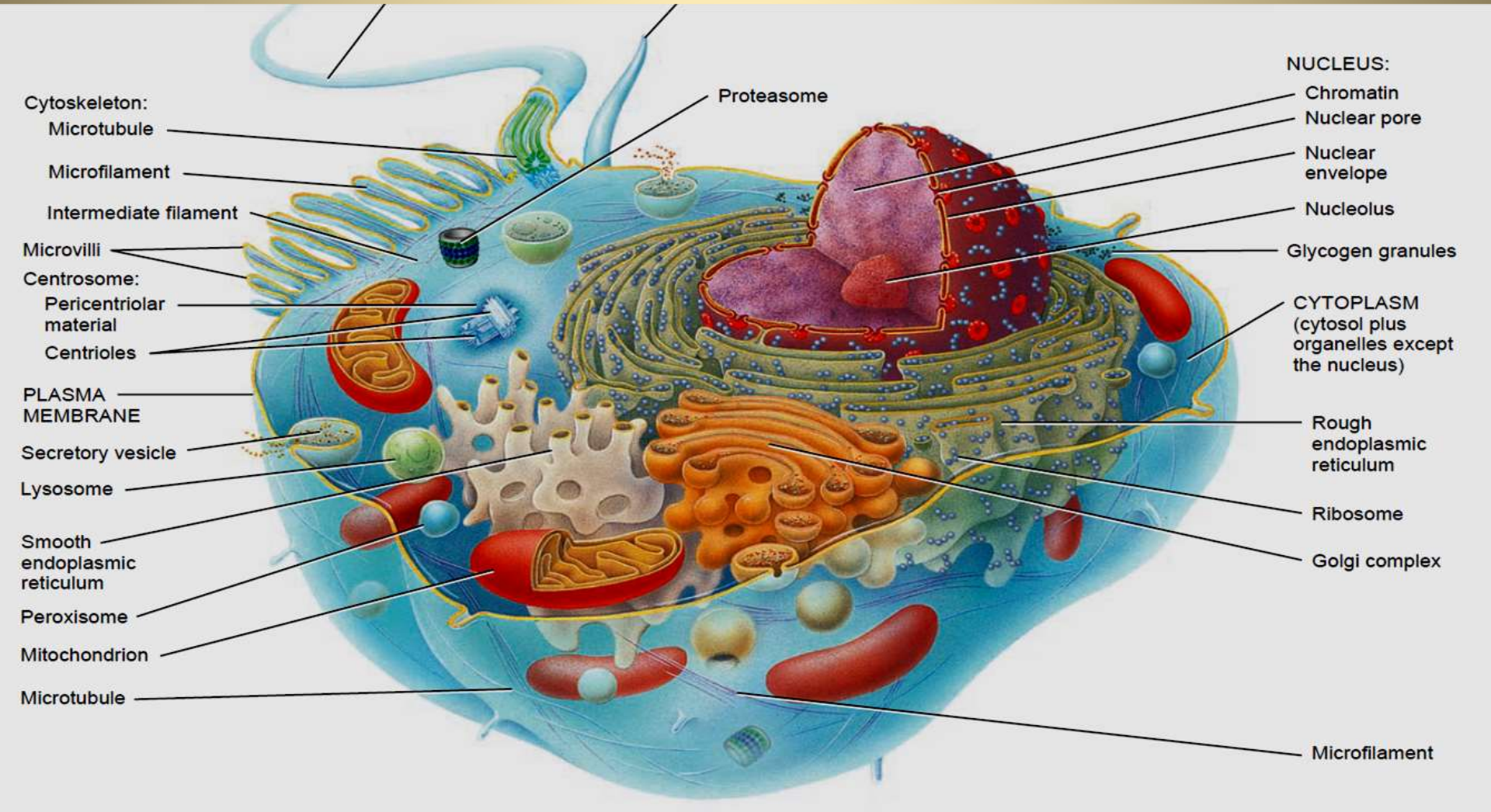
- Lacks a membrane bound nucleus
- Circular DNA , no histones
- Few internal structures
- Has a cell membrane (cell wall)
- Has ribosomes



Eukaryotic Cell

- Membrane bound Nucleus.
- Contains Cell Organelles.
- Linear DNA, Histones
- Unicellular to multicellular.

Structure of the Cell



Prokaryote Cell

1. Size – small 1- 10 µm
2. Unicellular
3. Has single membrane and cell wall
4. No nucleus.
5. Circular DNA
6. No Histones.
7. No cell organelles
8. Ribosomes – free in cytoplasm. 50S +30S {70S}
9. Cell division – fission
10. Cytoskeleton – absent

has flagella

Ex- bacteria,rickettsia

Eukaryote Cell

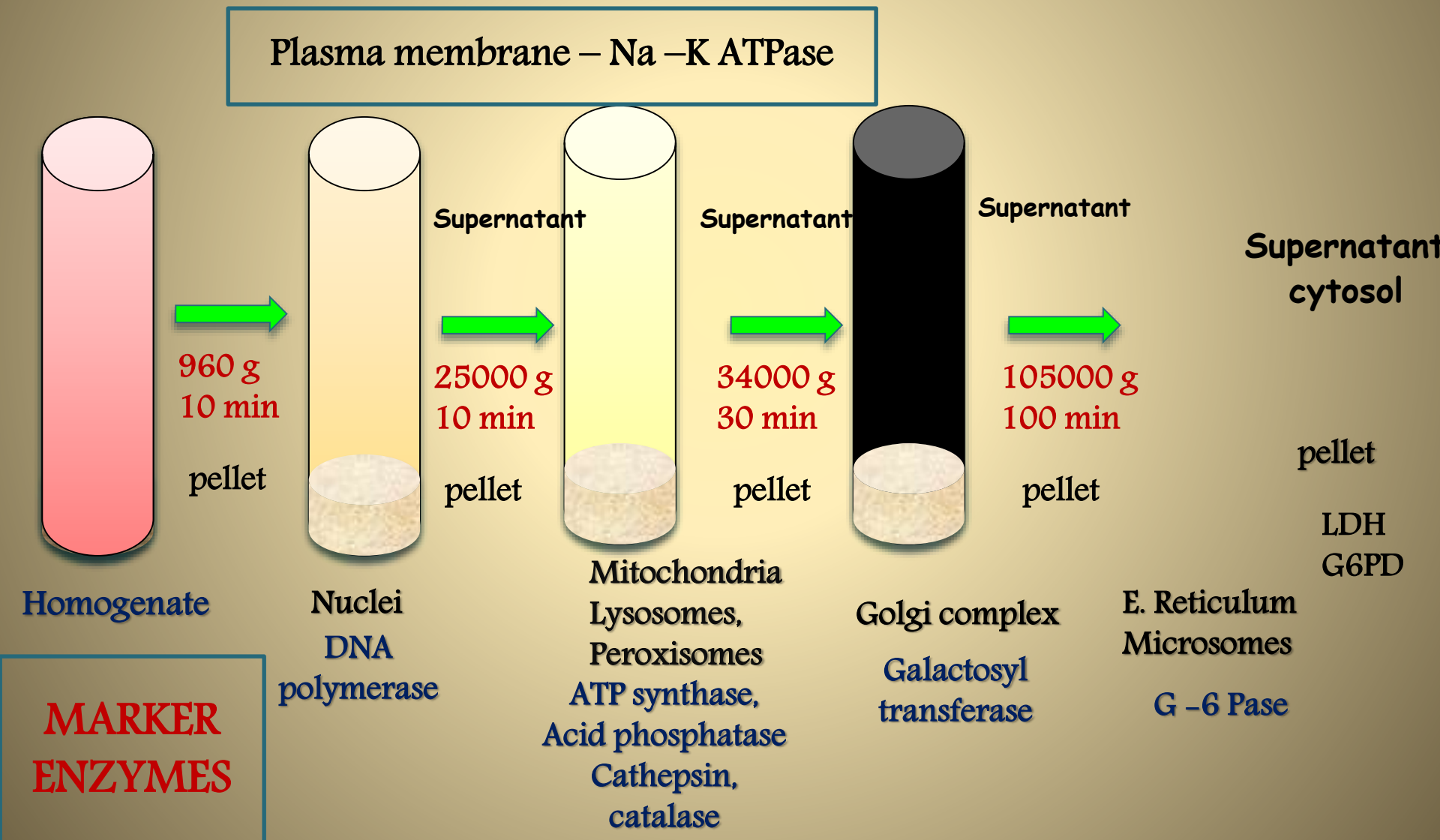
1. Large - 10 - 100 µm
2. Multicellular
3. Membrane bilayer.
4. Nucleus –well defined.
5. Linear DNA.
6. Histones
7. Membrane bound Organelles.
8. Ribosomes – on surface of E.R. 60S +40S {80S}
9. By mitosis.
10. Present.

Ex- fungi,plants,animals

Isolation of Subcellular Organelles

- Homogenization-
- By glass/teflon homogenizer, cells are disrupted by suspending in isotonic soln. (0.25 M sucrose, buffer at pH 7.4) and differential centrifugation.
- Isopyknic Centrifugation-
  - For organelles with same sedimentation coefficient - Separation based on their density.
  - Marker Enzymes- to assess purity of isolated subcellular organelle.

Isolation of Subcellular Organelles



- Plasma membrane
- Nucleus
- Cytoplasm
- Sub-cellular Organelles

Nucleus – control centre

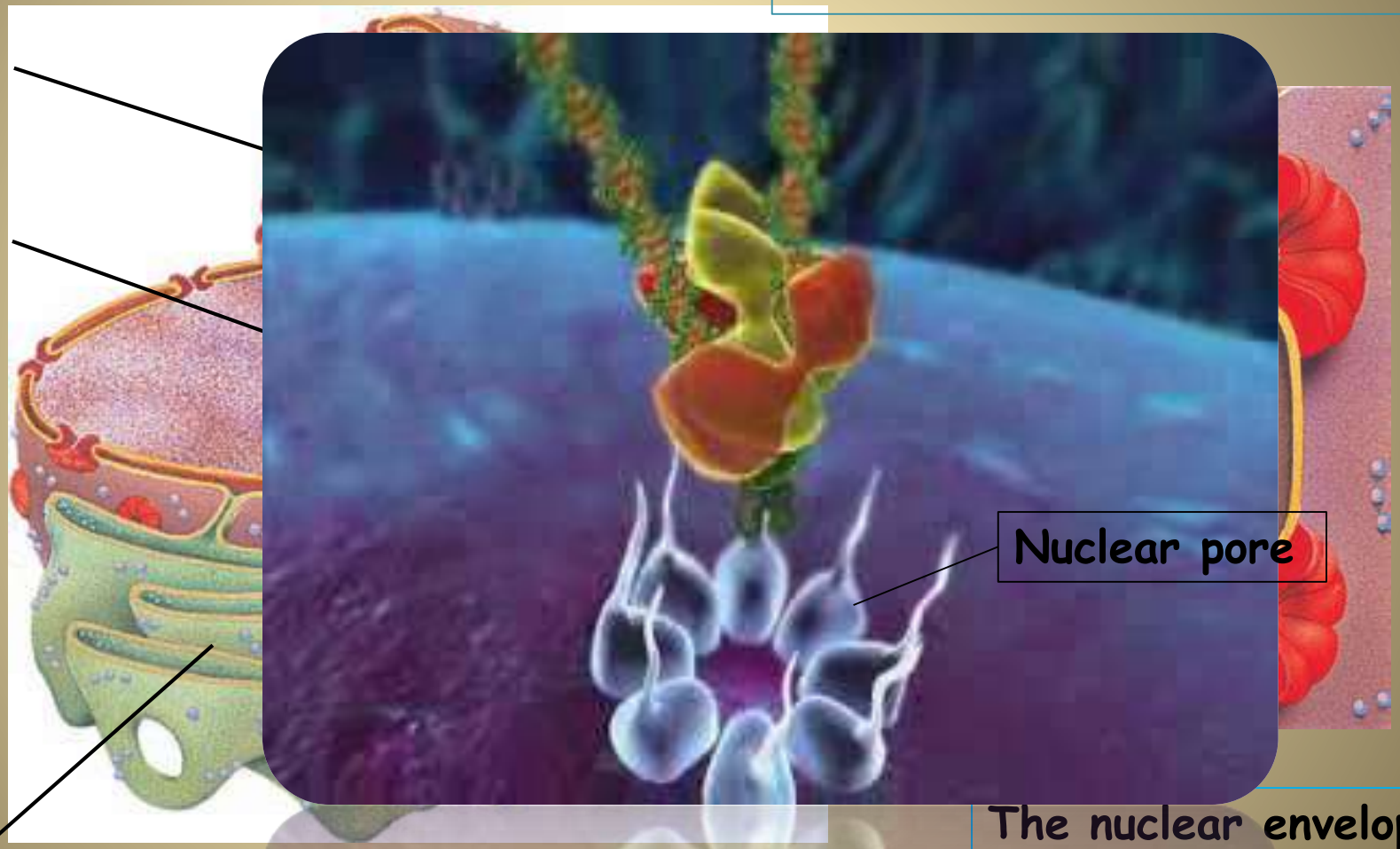
- Prominent organelle.
- Most cells have a single nucleus. Mature RBCs have none. skeletal muscle cells have multiple nuclei.
- Nuclear envelope – 2 membranes
  - outer – in continuity with E.R.
  - inner Peri nuclear membrane , with nuclear pores.
  - Nuclear pores - consists of a circular arrangement of proteins surrounding a large central opening
  - Control movement of proteins and RNA across envelope.

Nucleus – Information centre

- Contains DNA – chromatin.
- Nucleolus - dense body.
  - Ribosome Synthesis and r-RNA processing.
- Nucleoplasm- enzymes .ex-DNA Polymerase.
- Site for DNA Replication and RNA synthesis.

Nucleus

Functions
1. Contains DNA.
2. Directs cellular activities.
3. Produces ribosomes in nucleoli.

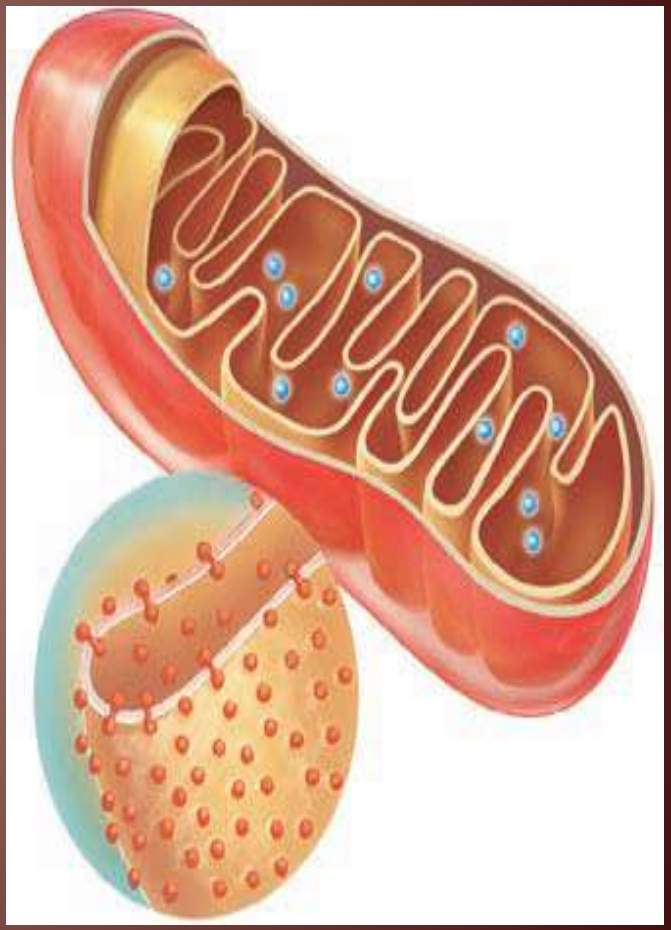


Details of the nucleus

The nuclear envelope

Mitochondria

- Elongated or rod shaped.
- Powerhouse of Cell.
- may be as few as 100 or as many as several 1000 depending on the activity of the cell.



- 2 membranes-
  - Outer – continuous
    - Lipid in nature.
    - Freely permeable (allows small molecules)
  - Inner – protein in nature .
    - High content of Cardiolipin.
    - impermeable.
    - folds to form cristae.(↑ surface area)
  - Inter-membrane space – adenylate kinase

Matrix- has specific Circular DNA , ribosomes and enzymes.

Mitochondrial DNA –maternally inherited.

Functions –

- Site of energy production – E.T.C.
- Site of metabolic pathways.- TCA cycle, Urea Cycle, F.A Oxidation etc.

Chloroplasts- in plant cells.

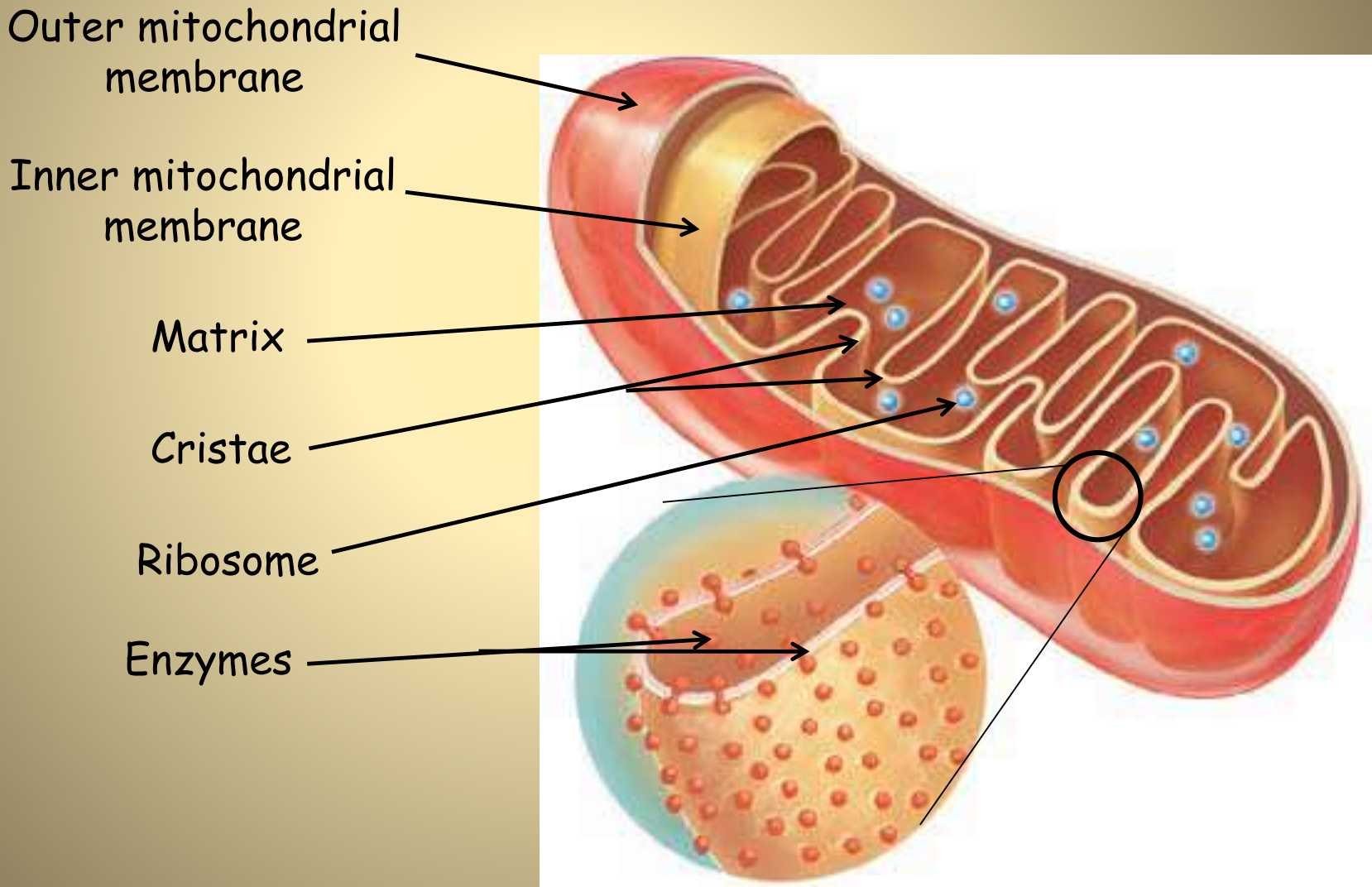
-Convert light energy into ATP by Photosynthesis.

Mitochondria

Function

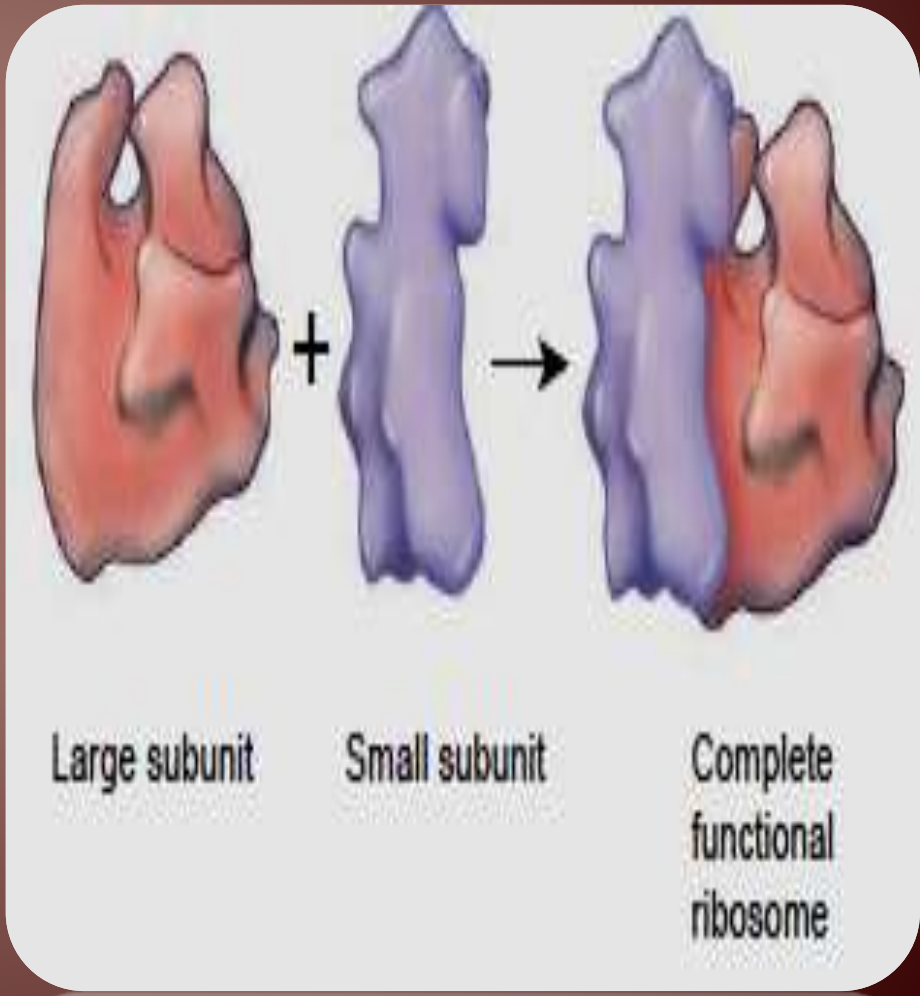
✓Generate ATP through reactions of ETC.

✓ Helps in Biochemical reactions. Such as oxidation of fatty acids, TCA, etc



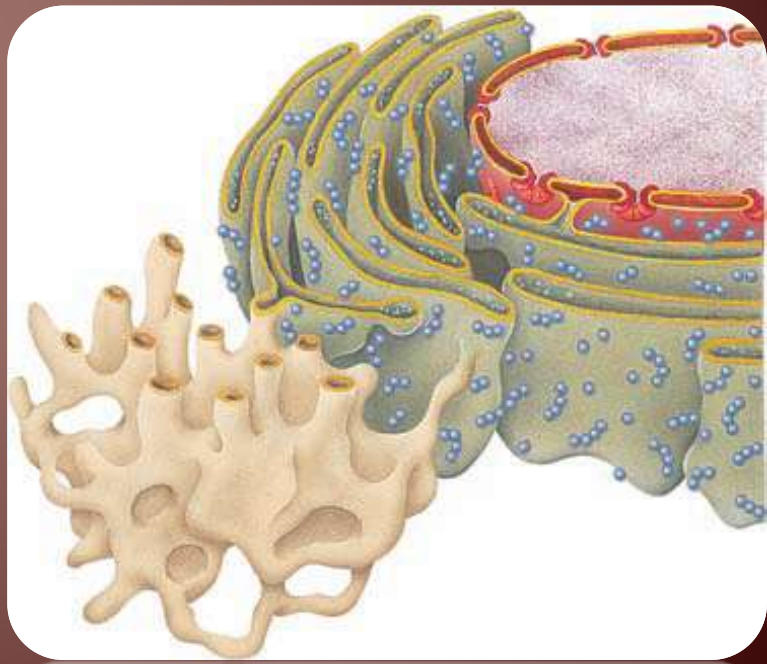
Ribosomes / protein factories

- Sites of protein synthesis.
- high content of r-RNA.
- They are present free in cytoplasm. or attached to RER consists of two subunits.
  - large subunit
  - small subunit.
- Prokaryotes- 70S- 50S +30S
- Eukaryotes- 80S-60S + 40S

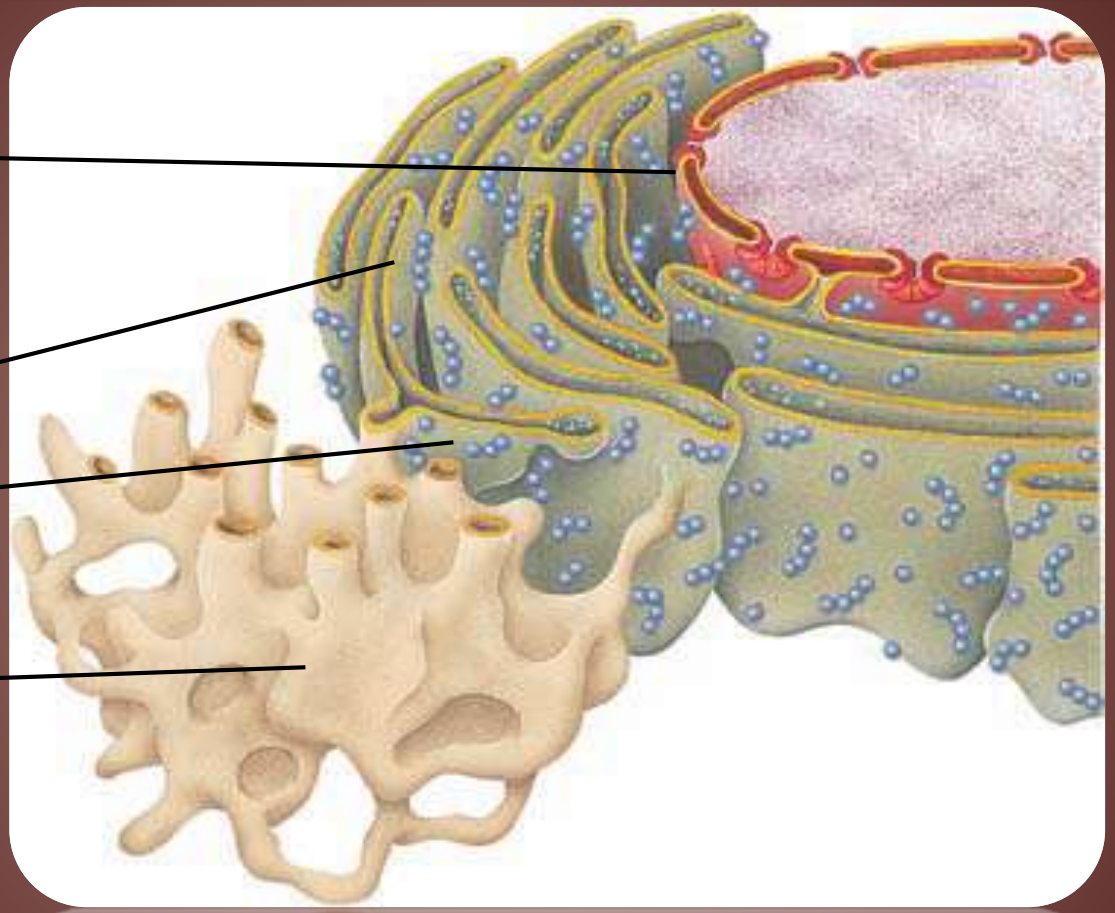


Endoplasmic Reticulum

- interconnected network of tubules and vesicles – cisternae
- Extend from nucleus to plasma membrane .
- Two types
  - -Rough ER
  - -Smooth ER



Structure of Endoplasmic Reticulum

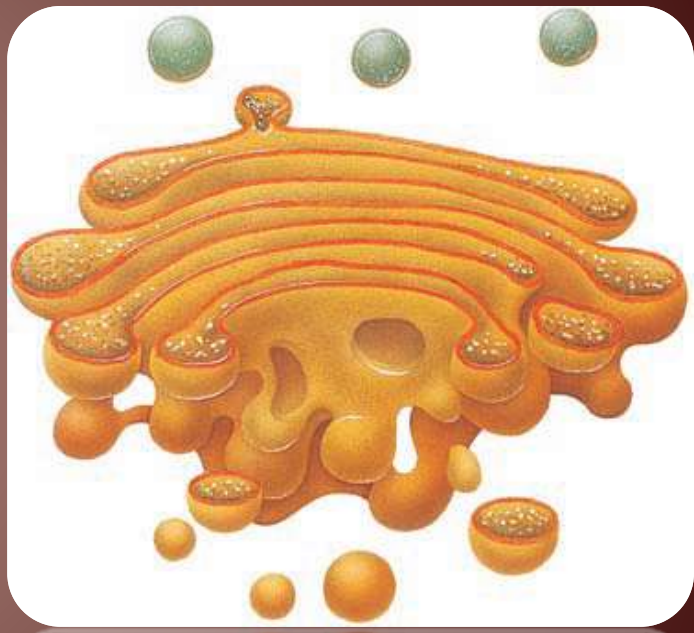


Functions

- RER- synthesizes proteins.
- SER- synthesizes phospholipids, cholesterol (in many tissue) & steroid hormones (adrenals, gonads).
- SER - site of Glycogen metabolism.
  - Removes the phosphate group from G-6-P; and release free glucose in blood.
- Sarcoplasmic reticulum - Stores & releases Calcium ions in the cells (that trigger contraction in muscle cells.)
- In liver & Kidney - Detoxifies drugs, toxins & Carcinogens.

Golgi Complex/Dictyosomes

- Consists of 3 to 20 cisternae, small, flattened membranous sacs.
- Prominent in cells that secrete proteins
- **SORTING UNIT-** Modifies, sorts, packages, & transports proteins from RER



Post-translational modifications.

E.R , Golgi – TRANSPORT NETWORK

Lysosomes





- Membrane-enclosed vesicles , from Golgi complex.
- Tiny organelles.
- SUICIDAL BAGS .
- 60 kinds of powerful digestive and hydrolytic enzymes.- Optimum pH – 5.
- Helps in fertilization.
- Role in Phagocytosis by W.B.C.
- Role in Cell Death- Autophagy
- Lysosomal Storage diseases – Genetic diseases, due to absent / deficient lysosomal enzymes. e.g, Niemann pick disease, Gaucher's disease.

Peroxisomes/ Microbodies

- Contain several oxidases. – Peroxidase , Catalase .

Functions :-

- Oxidation of amino acids.
- oxidation of long chain fatty acids.
- Protects cell from the toxic effects of H_2O_2.
- Dysfunction of Peroxisomes leads to **Zellweger syndrome**
- **GLYOXSOMES-** in plants,
  - In seeds –rich in lipids. (fats – succinate)

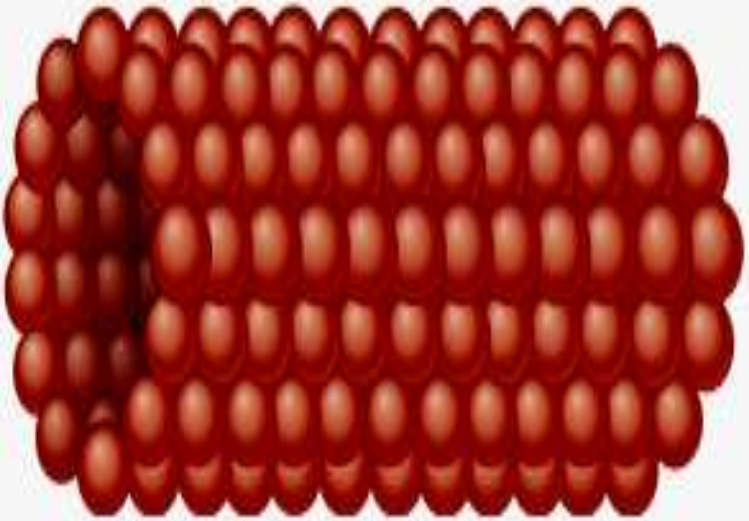
Cytoplasm

- Fluid content of cell.
- Site for many metabolic pathways.

ex- Glycolysis, Protein synthesis , fatty acid synthesis, purine synthesis.

Cytoskeleton

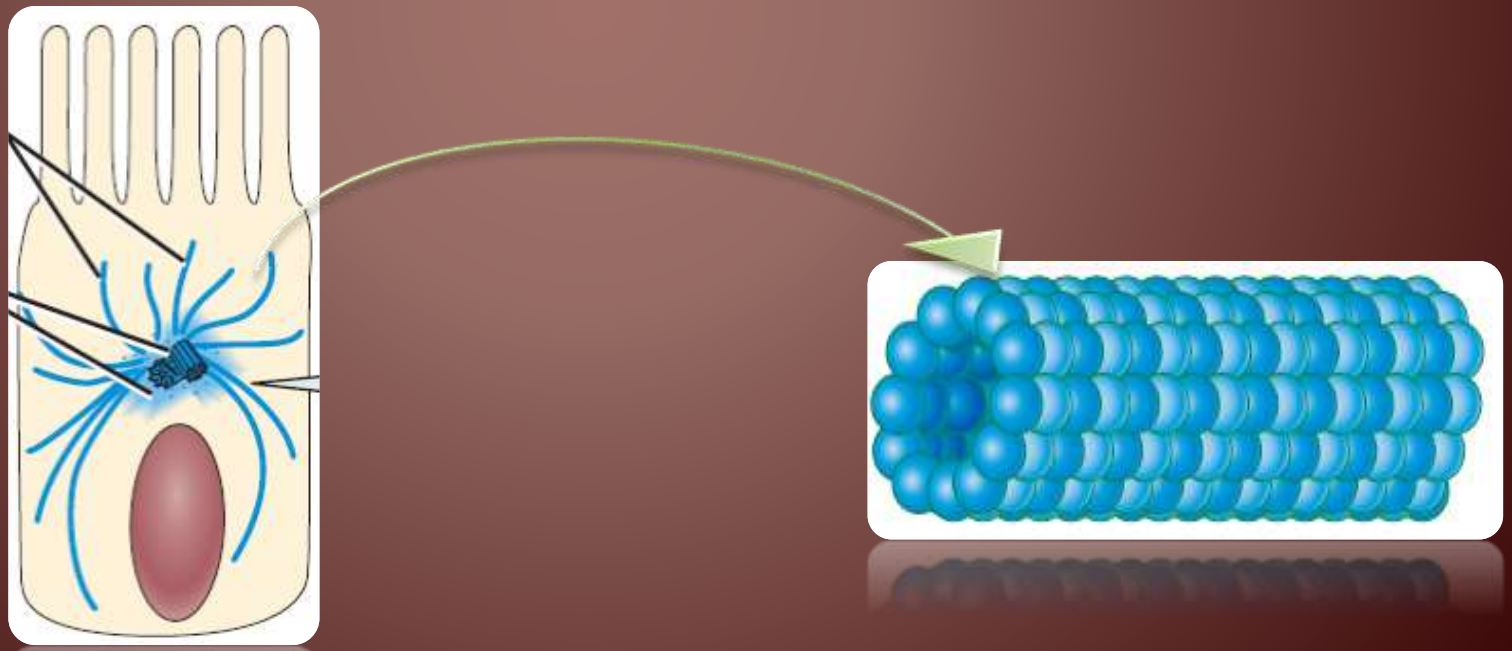
- A network of protein filaments that extends throughout the cytoplasm.
- anchored to plasma membrane.
- Dynamic structure.
- Three types of filamentous proteins -
  - Microtubules
  - Intermediate filaments
  - Microfilaments
- Provides shape to cell. Acts as internal framework.
- Helps in uptake of materials into cell.
- Helps in internal movement of cell organelles , movement of cells and muscle contraction.
- Helps in Cell division.

	Cytoskeletal filaments	Diameter (nm)	Protein subunit
	Microfilament	7	Actin
	Intermediate filament	10	Several proteins
	Microtubule	25	Tubulin

Source: Barrett KE, Barman SM, Boitano S, Brooks H: *Ganong's Review of Medical*

Microtubules

- Long, Hollow, unbranched, polar cylinders.
- made up of protein “tubulin”(α & β tubulin),
- The largest of cytoskeletal components.
- Major components of axons and dendrites.

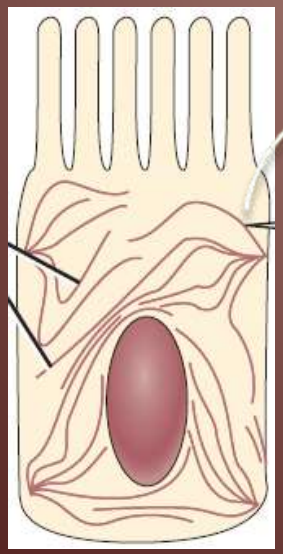


Microtubules

- Functions:-
- Microtubules help in structural support and maintain the shape of the cell.
- Helps in movement of organelles,secretory vesicles and exocytosis.
- Formation and function of mitotic spindle.
- Movement of cilia and flagella.
- Disorder:-
- Primary ciliary dyskinesia- associated with celiary dysfunction
- clinical effect- recurrent upper & lower resp. tract infection, male infertility

INTERMEDIATE FILAMENTS

- Polymers of long rod like proteins.
- These filaments are thicker than microfilaments but thinner than microtubules.
- Made up of – Keratin,Desmin,Neurofilaments etc.





Functions :-

- Provide mechanical support to the cell.
- Help in intercellular attachment.
- Provide strength and rigidty to neurons.
- Major structural role in skin and hair cells.

Microfilaments

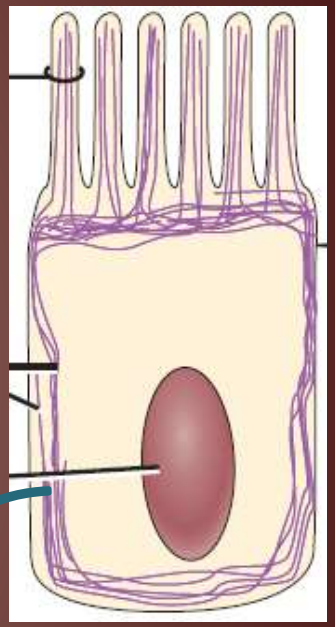
- Thinnest elements of the cytoskeleton.
- Composed of the protein actin, (β , γ actin)
- Form a meshwork under plasma membrane
- Stress fibres.

Functions :-

- Mechanical support for the basic strength & shapes of cells.

ex- Microvilli is rich in microfilaments – Shape.

- involved in muscle contraction, cell division, and cell locomotion.



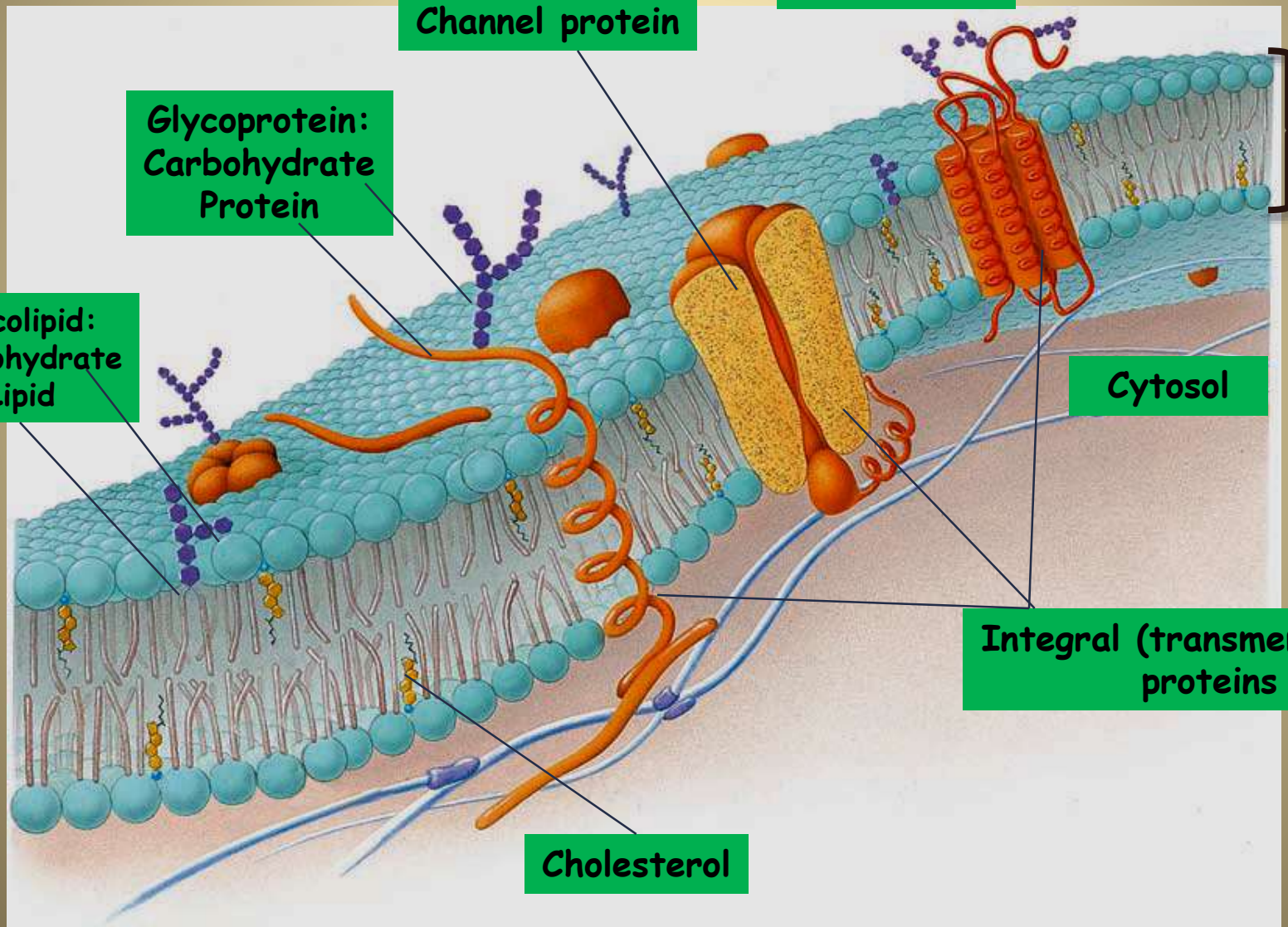


- Mitochondria-ATP production , metabolic pathways
- Nucleus-contains genetic material
- Ribosome-assembles proteins
- Endoplasmic Reticulum-protein translation, folding and transport
- Golgi Apparatus-delivery system for the cell maturation of proteins.
- Transport network- E.R ,Golgi.
- Chloroplast-conduct photosynthesis
- Vacuole-Storage, secretory, excretory
- Lysosomes -digestion, cell death,
- Peroxisome – breaks toxic substances.
- Glyoxysome -breakdown of fatty acids to sugars

PLASMA MEMBRANE

- Selectively permeable barrier that surrounds the cytoplasm of a cell.
- Lipid bilayer – Davson & Danielle.
- Is described by the fluid mosaic model- Singer and Nicolson.
- Made up of 3 macromolecules……
  - Lipids
  - Proteins
  - Carbohydrates.

Structure of Plasma membrane



Membrane Lipids

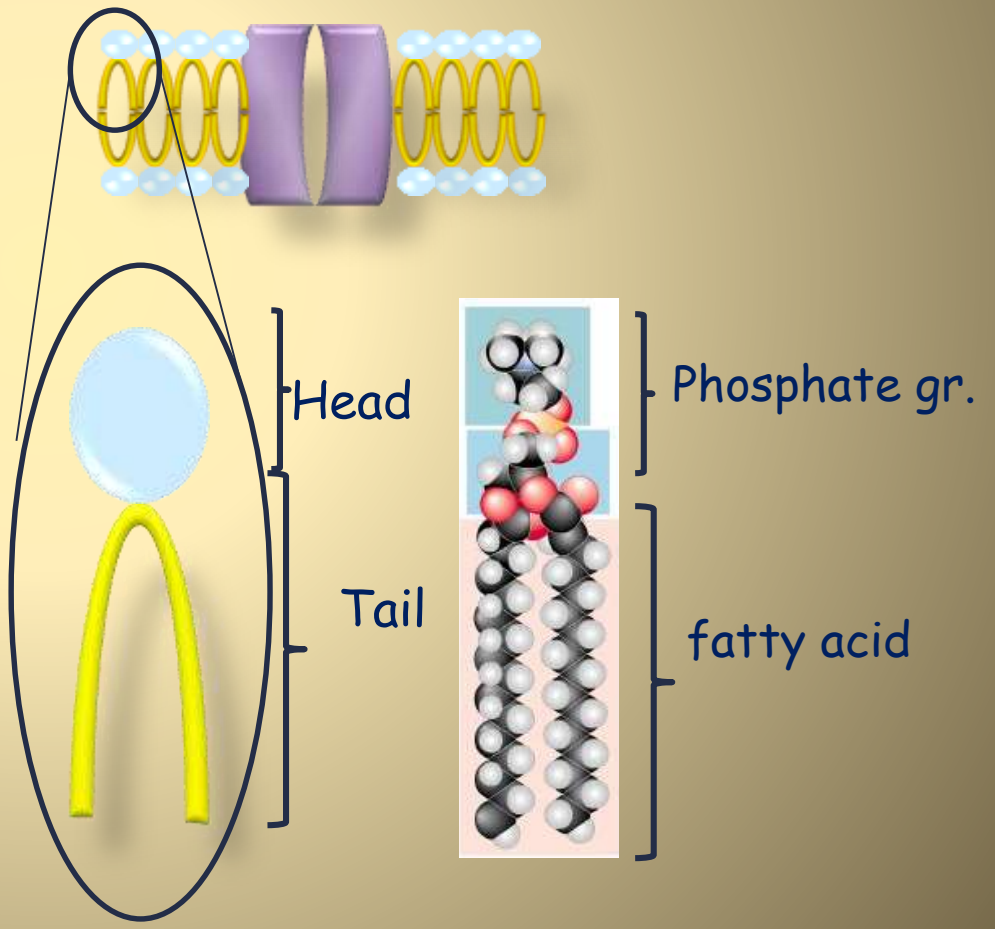
- 75% - Phospholipids
- 20% - Cholesterol
- 5% - Glycolipids.

- Most of them are - Amphipathic (polar & nonpolar)
- Acts as Permeability barriers.
- Essential for the maintenance of fluidity of membranes.

Phospholipids

- "Head" – Polar part – phosphate group
- "Tail" – Non polar part – long chain fatty acids
- include……

  Glycero, Sphingo P.L – phosphptidylcholine, phosphptidylinositol, plasmalogens & sphingomyelin.
- These are not linked to neighbouring P.L by any chemical bonds – lateral movements



Cholesterol –

- weakly amphipathic
- interspersed among other lipids in both layers of the membrane.
- Stability to membrane .
- alters Fluidity of membrane.

Fatty Acids-

- unsaturated cis fatty acids - ↑ fluidity.

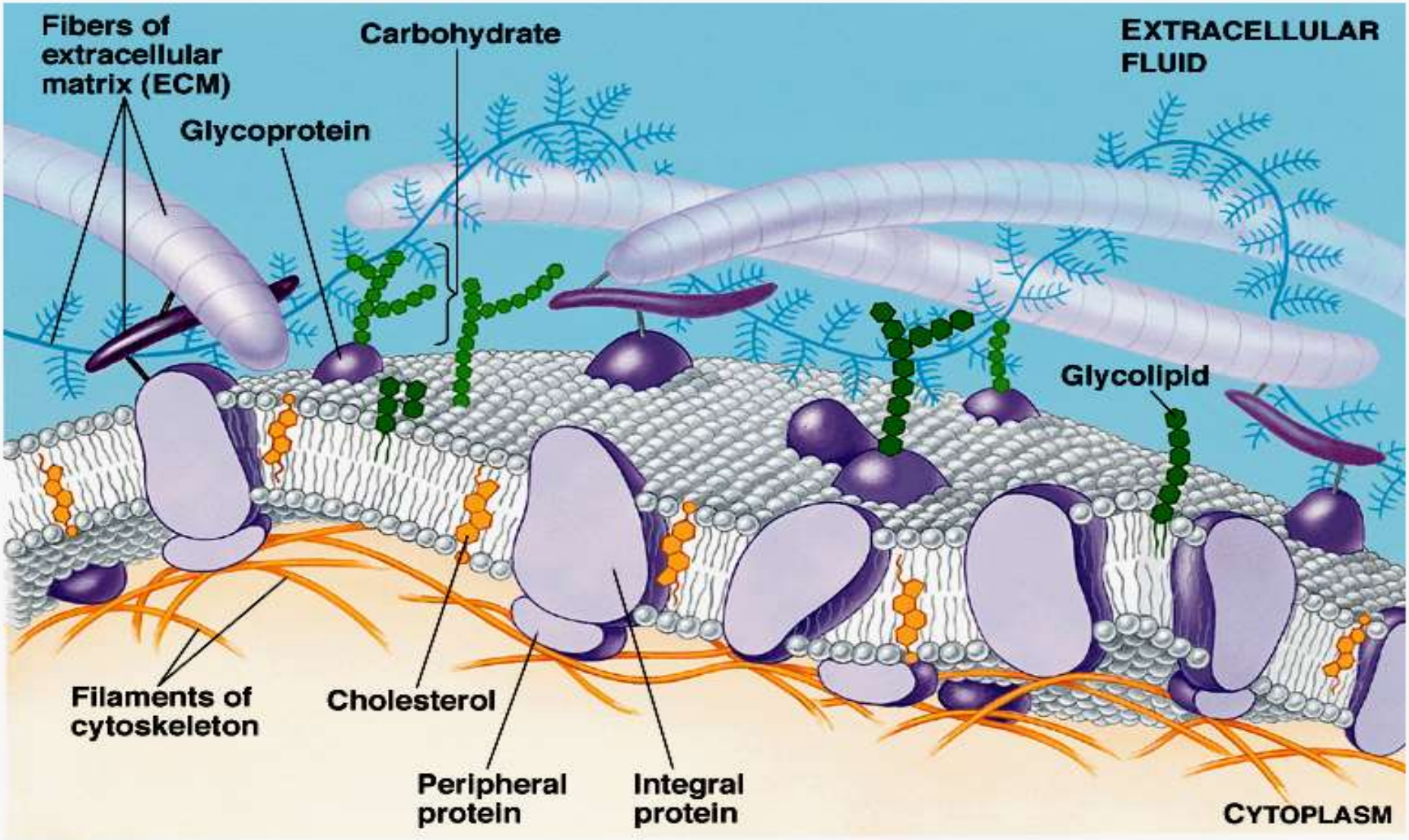
Glycolipids -

- present only on the outer surface of membrane.

Membrane Proteins -

- Two types of proteins are present in membrane -
  - Integral proteins
  - Peripheral proteins
- Integral proteins - partially / totally immersed in it.
- Most integral proteins are *transmembrane proteins-* extend through out the lipid bilayer.
- Most of them are glycoproteins.
- Peripheral proteins - bind loosely with the polar heads of membrane lipids at the inner or outer surface of the membrane.

Membrane Proteins



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Functions

Transmembrane proteins –

- Ion channels
- Carriers (transporters)

Peripheral proteins –

- Receptors
- Enzymes

Membrane Carbohydrates

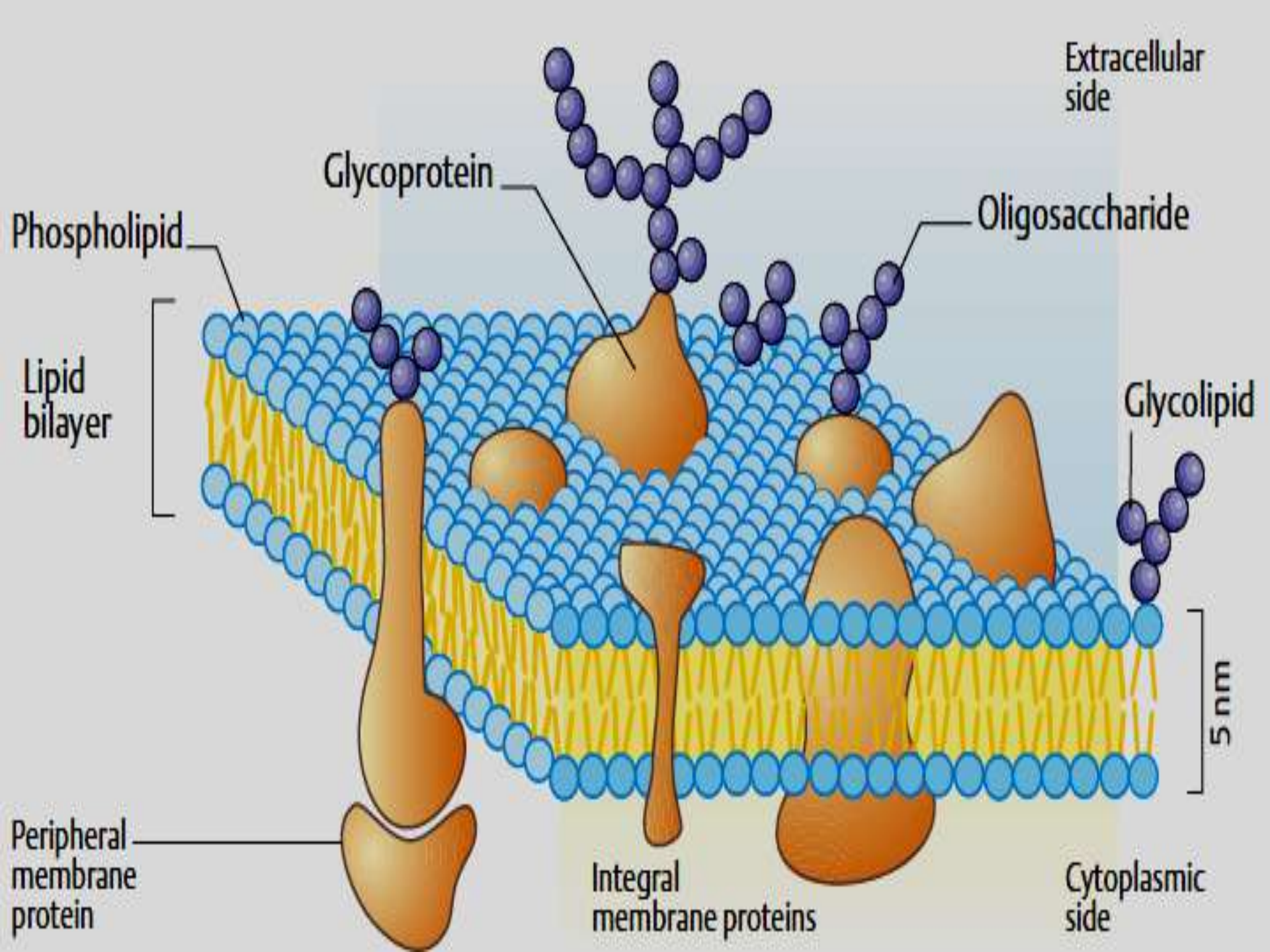
- covalently bound to lipids and proteins to form glycolipids and glycoproteins.
- These are mostly - Glucose , Galactose , Mannose
  - N-acetyl glucosamine , N-acetylgalactosamine
- Proteoglycans- outer surface.
- Glycocalyx – loose CHO layer on outer surface of cell .

Functions –

- impart –ve charge to cell- repels other particles.
- helps in inter-cellular attachment .
- act as receptors.
- Cell identity markers (glycoproteins & glycolipids) , antibody processing.

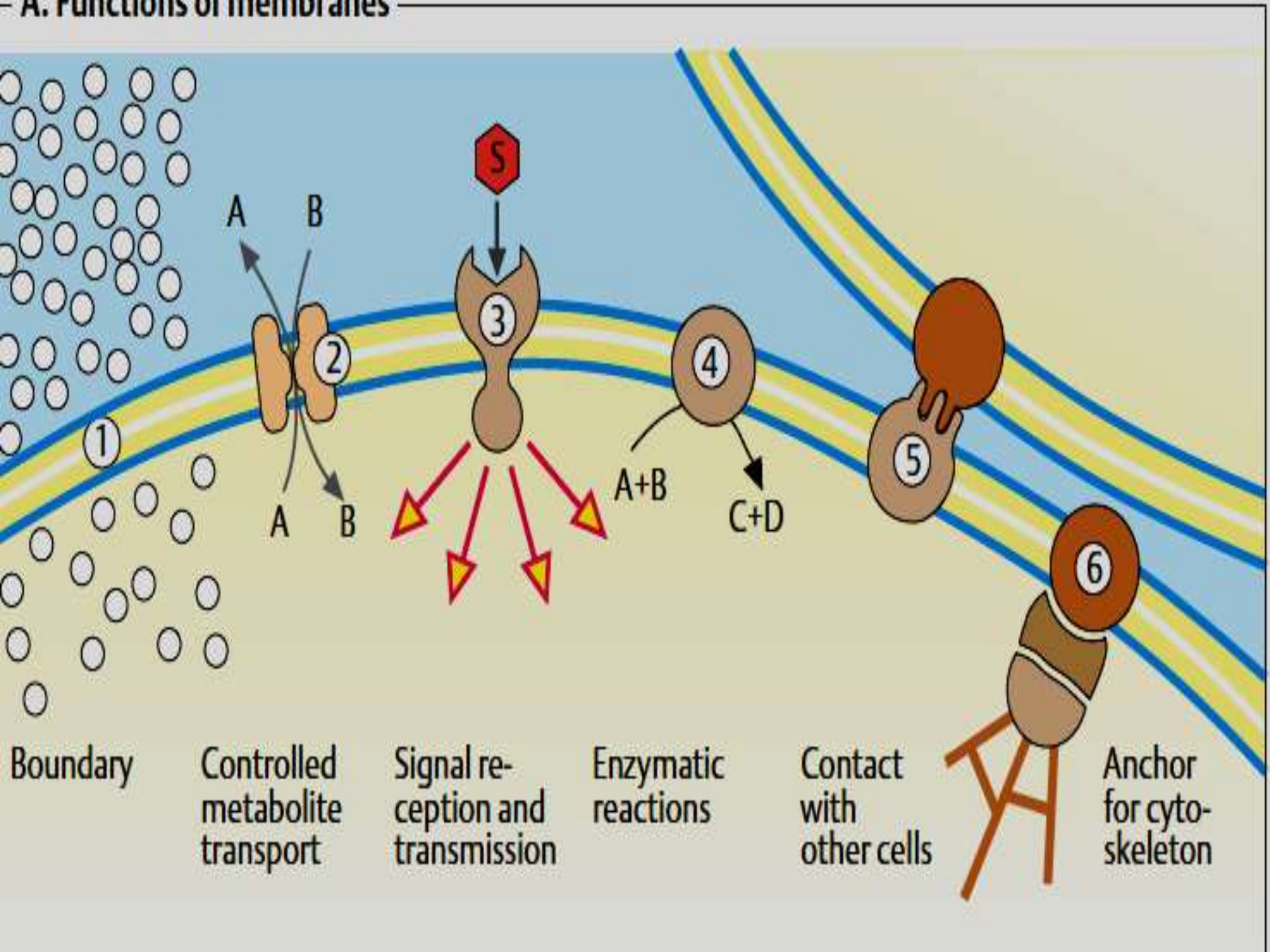
Special features –

- **Fluid mosaic model – in a sea of lipid bilayer , proteins float and arranged in a mosaic like pattern.**
- **Asymmetric – outer and inner face of membrane have different components.**
- **Fluid in nature – US.F.A bound to P.L - fluidity of membrane, which aids in function.**
- **Anchored to Cytoskeleton .**


Extracellular side
Glycoprotein
Oligosaccharide
Phospholipid
Lipid bilayer
Glycolipid
5 nm
Peripheral membrane protein
Integral membrane proteins
Cytoplasmic side

Functions of cell membrane

- **Acts as a semi-permeable barrier.**
- **Associated with several enzymes.**
- **Contain receptors for hormones.**
- **Contain recognition sites for antibodies.**


A. Functions of membranes
S
A
B
A
B
A+B
C+D
1
2
3
4
5
6
Boundary
Controlled metabolite transport
Signal re-ception and transmission
Enzymatic reactions
Contact with other cells
Anchor for cyto-skeleton

Transport across Cell membrane

- Essential to maintain equilibrium of cell
- Certain substances must move into the cell to support metabolic reactions.
- Other substances produced by the cell for export

  or as cellular waste products must move out of the cell.

Transport across Cell membrane

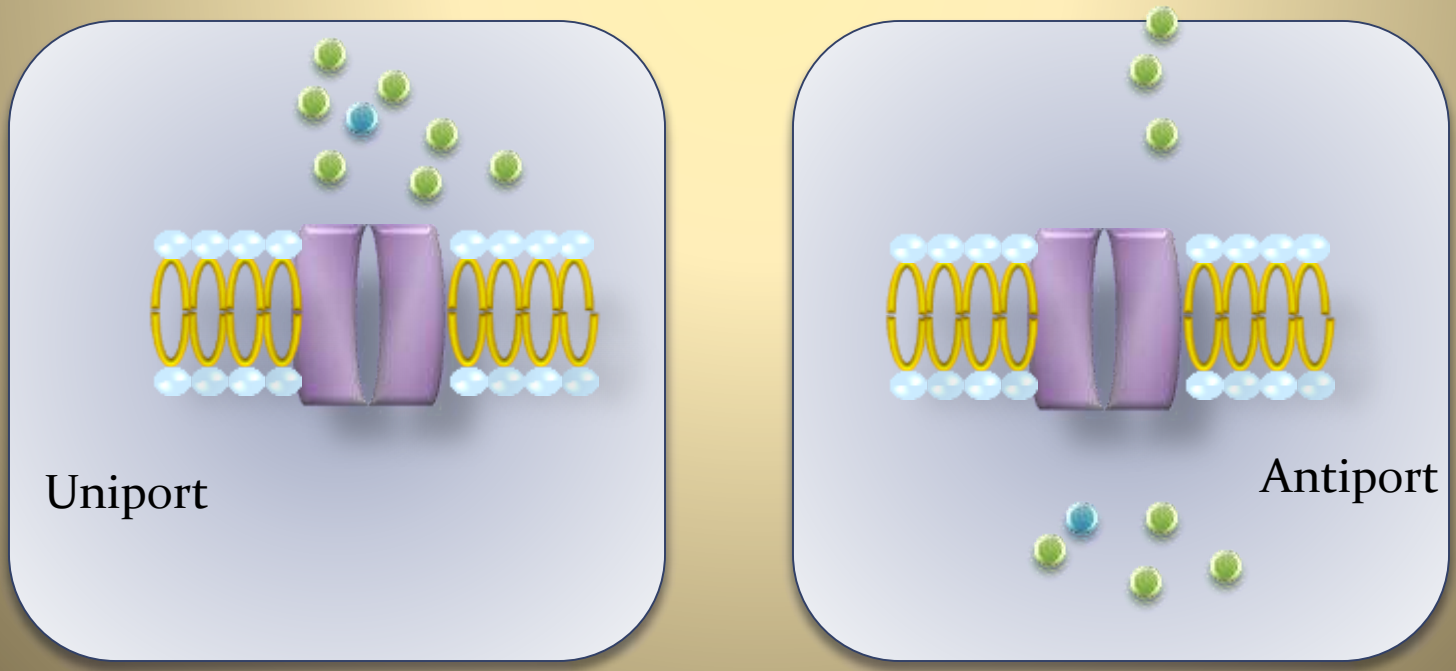
- Uniport
- Cotransport

Mechanisms –

- **Passive transport.**
  - Simple diffusion
  - Facilitated diffusion
- **Active transport.**
  - Primary Active Transport
  - Secondary Active Transport
- **Bulk Transport [MACROMOLECULES]**
  - Exocytosis.
  - Endocytosis.

- Uniport- Transport of single type of molecule in one direction. E.g. transport of glucose in RBC by GLUT.
- Cotransport-
  - Symport – Transport of molecules in same direction.

    ex- Na- glucose transporter.
  - Antiport - Transport of molecule in opposite direction. E.g. chloride-bicarbonate exchanger.



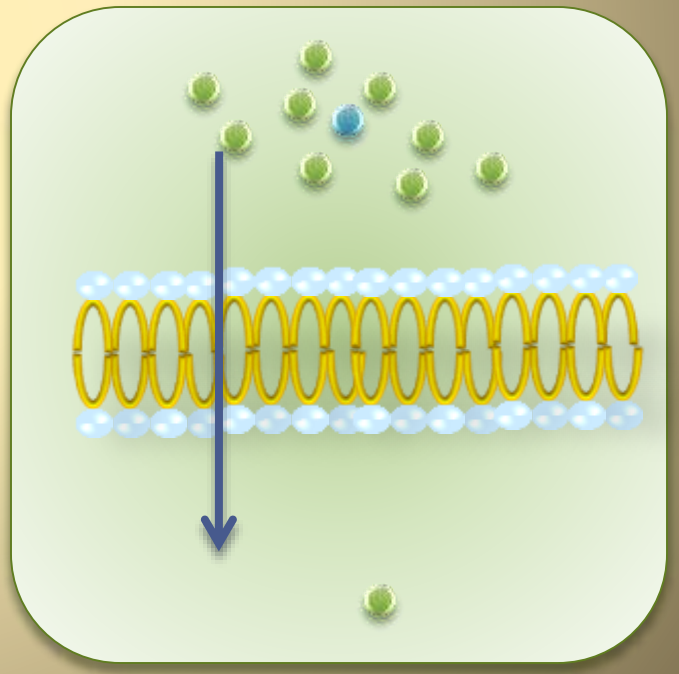
Passive Transport

Simple Diffusion :-

- Movement of particles from the area of higher conc. to an area of lower conc. (i.e., along the conc. gradient).
- It does not require energy and carrier proteins.

Ex. – Transport of -

- gases
- neutral polar molecules
- lipid soluble molecules



Channels- Diffusion of ions through membrane.

- Protein in nature.
- selective .
- Moves from high conc. to low conc.

Filters – allows only one ion.

Ex- K. channel allows K ion 100 times more than Na ion.

Osmosis.

- The diffusion of water through a semipermeable membrane.
- Movement of water molecules occur from an area of lower solute concentration to an area of higher solute concentration.
- Clinical Significance-
  - ✓ Decreased formation of urine in hypovolemic conditions.
  - ✓ Edema due to hypoalbuminemia.
  - ✓ Tonicity and its effects on red blood cells (RBCs).



(a) Illustrations showing direction of water movement

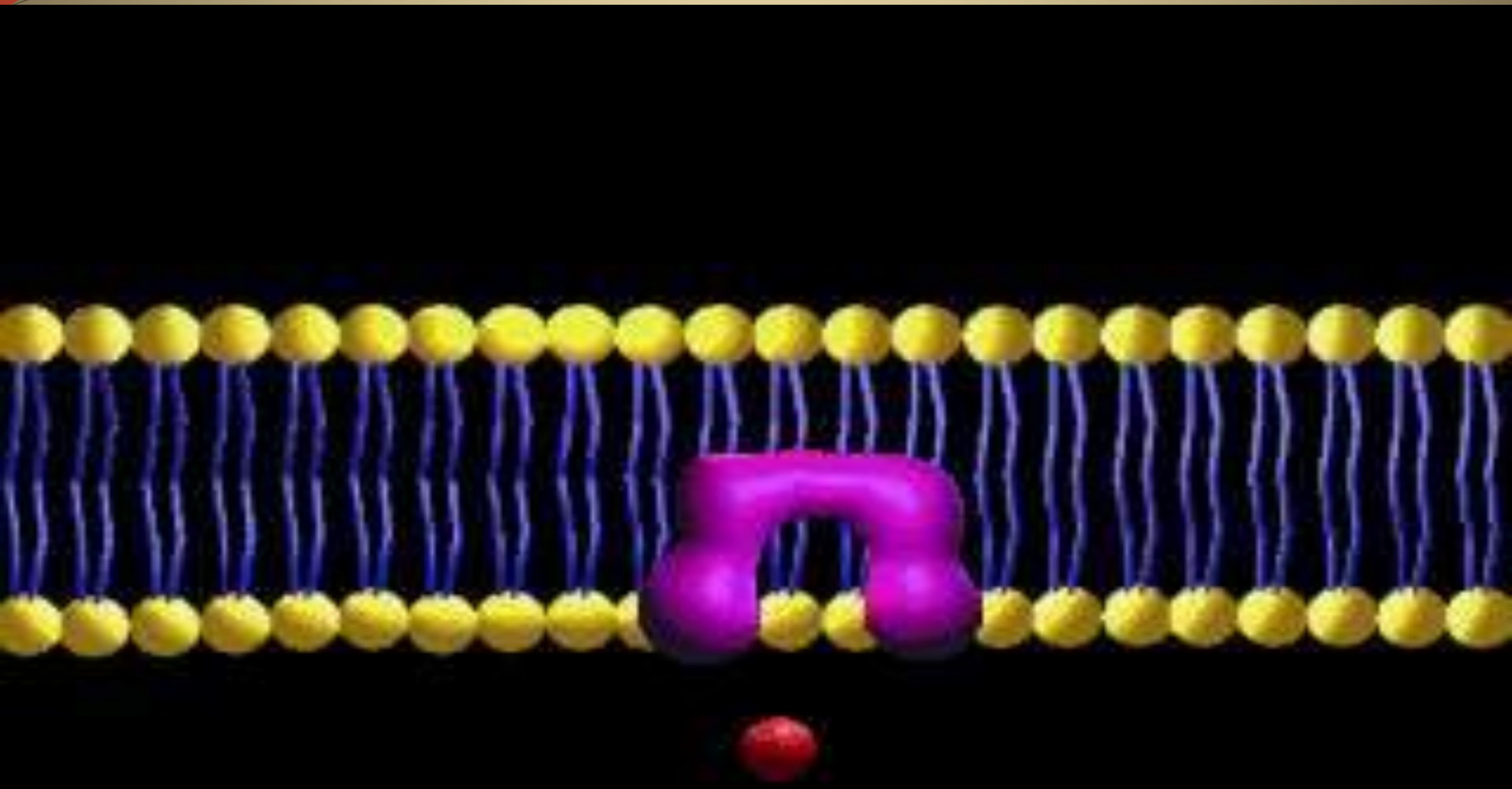


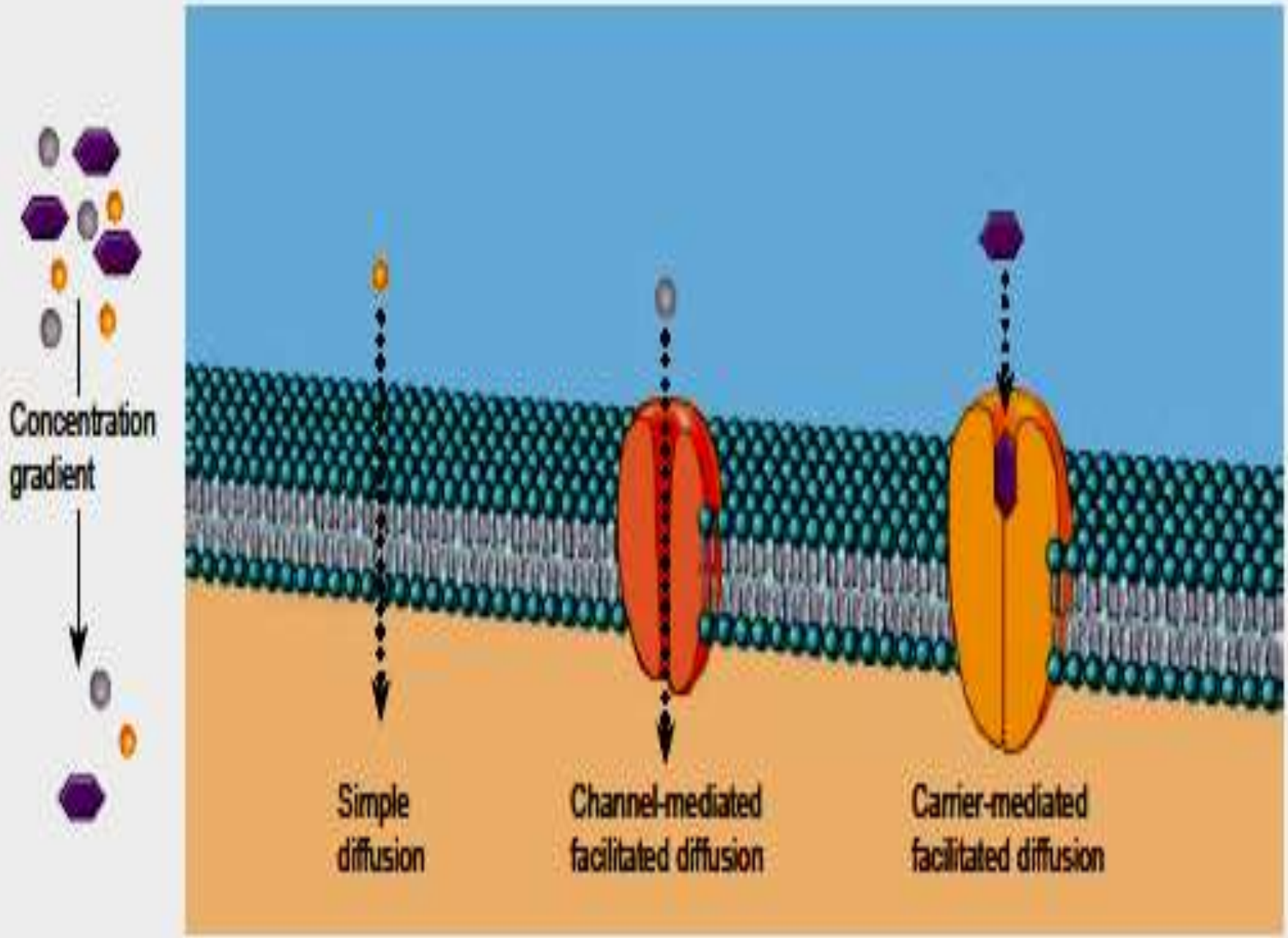
Normal RBC shape

RBC undergoes hemolysis

RBC undergoes crenation

Facilitated diffusion-




Concentration gradient
Simple diffusion
Channel-mediated facilitated diffusion
Carrier-mediated facilitated diffusion

Active Transport.

- **Two types**
  - **Primary Active Transport.**
  - **Secondary Active Transport.**

Primary Active Transport.

- Transport against conc. Gradient .
- carrier mediated.
- requires energy.
  - used directly from hydrolysis of ATP .
- Ex. – sodium-potassium pump,
  - calcium pump / ca^{+2}- ATP ase .

Sodium-Potassium Pump (Na^+ - K^+ATPase)

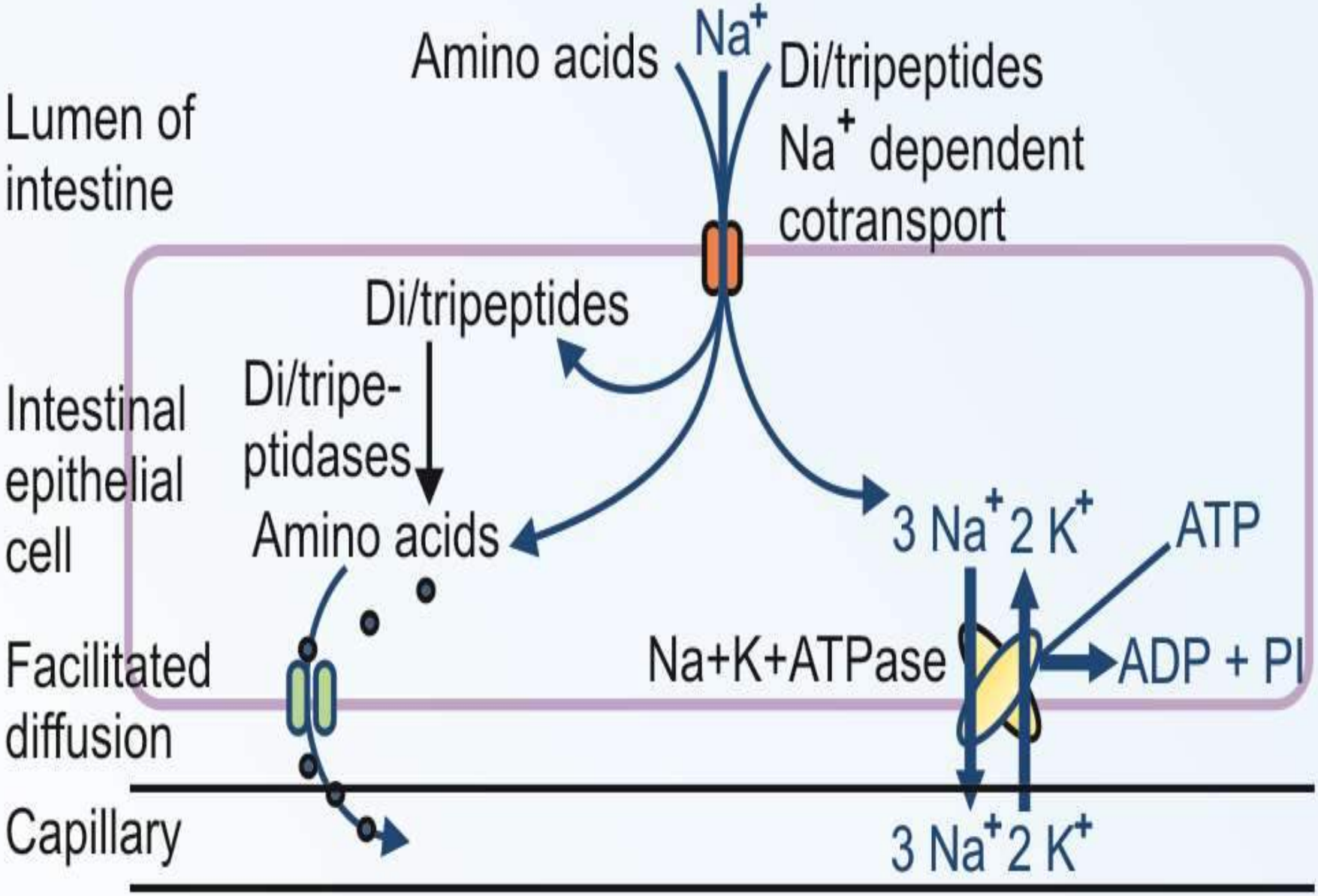
- Also called Sodium pump.
- In this pump, 3 sodium ions move out of the cells and 2 potassium ions move inside the cell, with consumption of 1 ATP molecule.
- This is to maintain –
  - low intracellular Na^+ & high intracellular K^+ - generating an electrochemical gradient.
  - nerve and muscle cell excitability
  - active transport of sugars and aminoacids.
- Ca $^{+2}$ pump- maintains low ca^{+2} in cell and high ca^{+2} in sarcoplasmic reticulum.

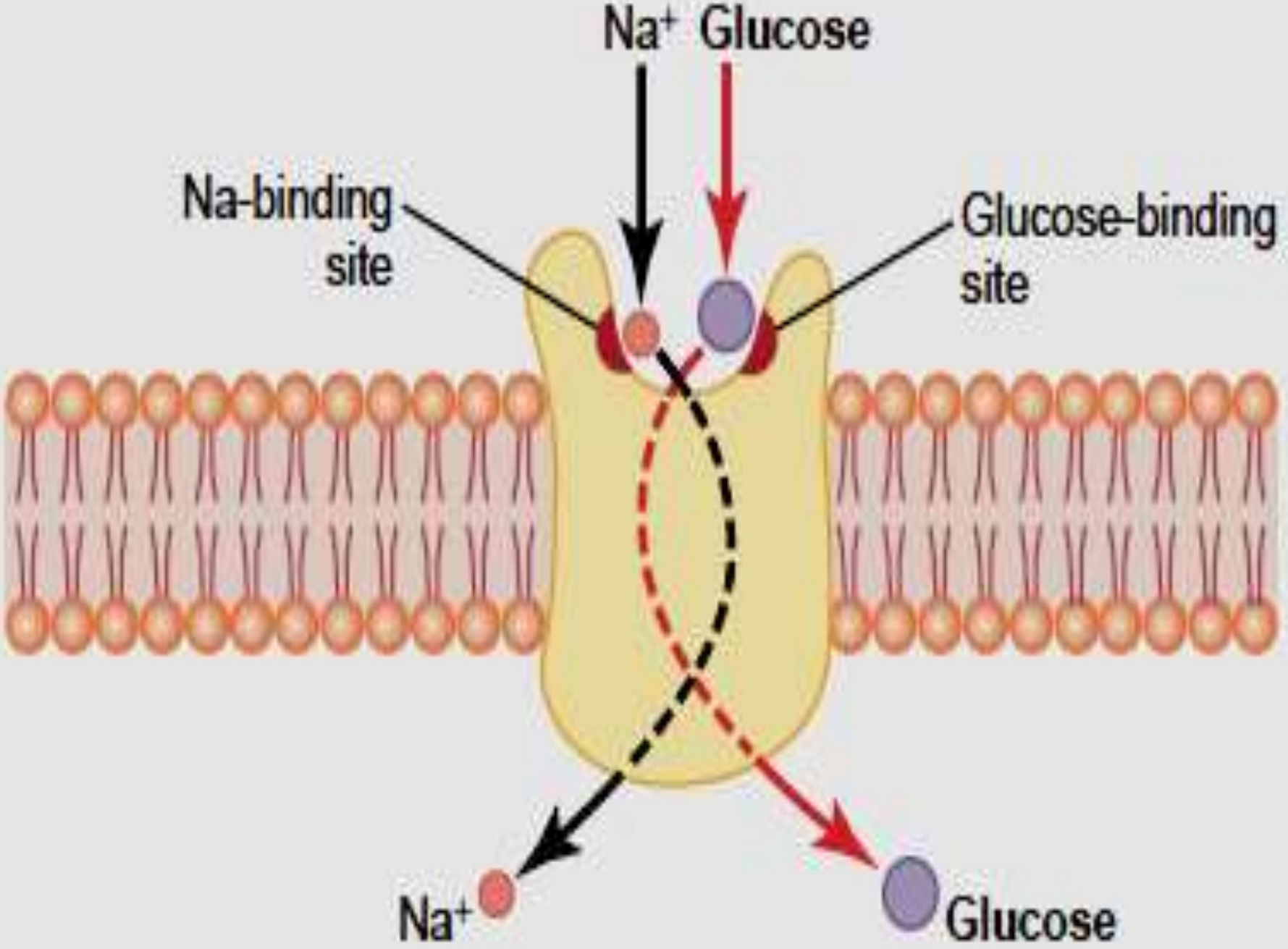
Secondary Active Transport.

- Transport against conc. gradient in which energy is used indirectly.
- The transport of two or three molecules are coupled.
- This transport is coupled with Na-K ATPase, that requires the ATP.
- It occurs by symport and antiport.

Symport (co-transport)

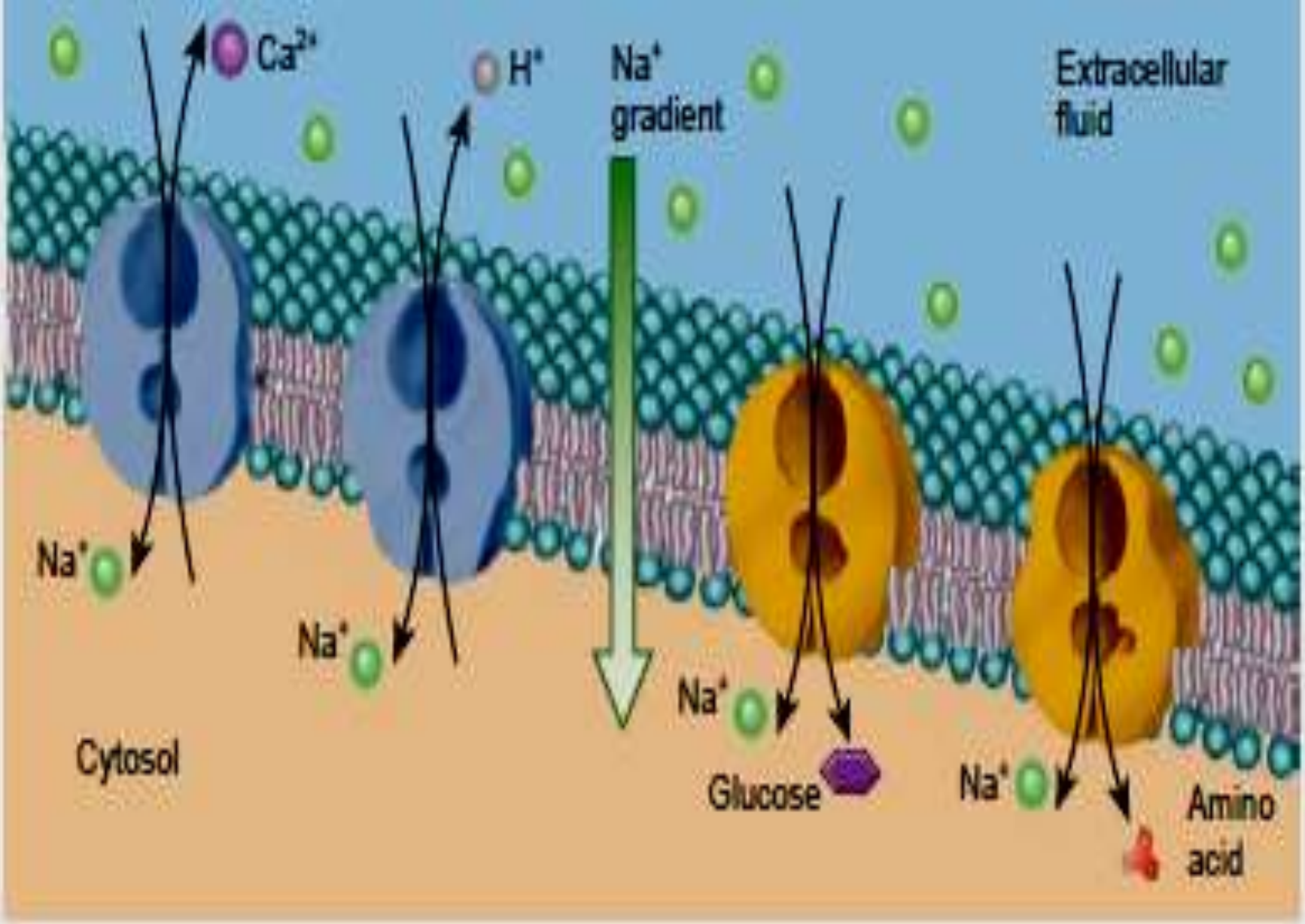
- Molecule move in same direction.
- Example-
  - Sodium-glucose symport
  - Sodium-amino acid symport.
- Disorders- cystinuria & Hartnup disease (mutations in sod-amino acid symport)


Amino acids
Na+
Di/tripeptides
Na+ dependent
cotransport
Lumen of
intestine
Di/tripeptides
Di/tripe-
ptidases
Intestinal
epithelial
cell
Amino acids
3 Na+ 2 K+
ATP
Na+K+ATPase
ADP + PI
Facilitated
diffusion
Capillary
3 Na+ 2 K+


Na+ Glucose
Na-binding site
Glucose-binding site
Na+
Glucose

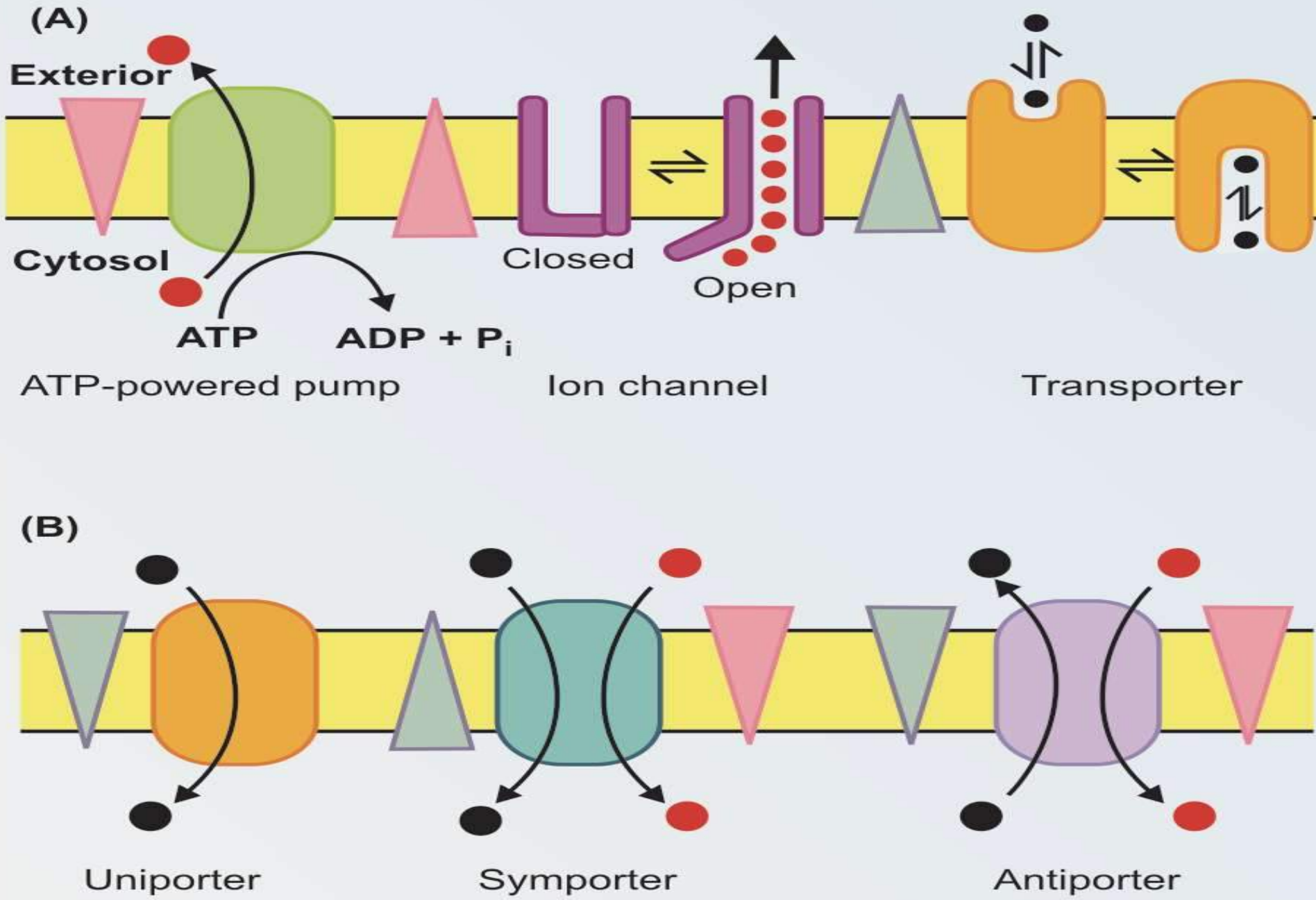
Antiport (counter transport)

- Molecule move in opposite directions.
- Example-
  - Sodium-hydrogen exchanger (Renal tubule)
  - Calcium-hydrogen exchanger (cardiac muscle)



(a) Antiporters

(b) Symporters


(A)
Exterior
Cytosol
ATP
ADP + Pi
Closed
Open
ATP-powered pump
Ion channel
Transporter
(B)
Uniporter
Symporter
Antiporter

Bulk transport

- involves transport by formation of membrane bound vesicles.
- involves transport of macromolecule.
- Requires energy-ATP , Ca^{+2} ions.

Endocytosis

- Engulfing large molecules by the cell.
- Two type of endocytosis.
  - Phagocytosis-cell eating

    ingestion of large molecules, such as bacteria into the cell.

    - occurs only in specialised cells

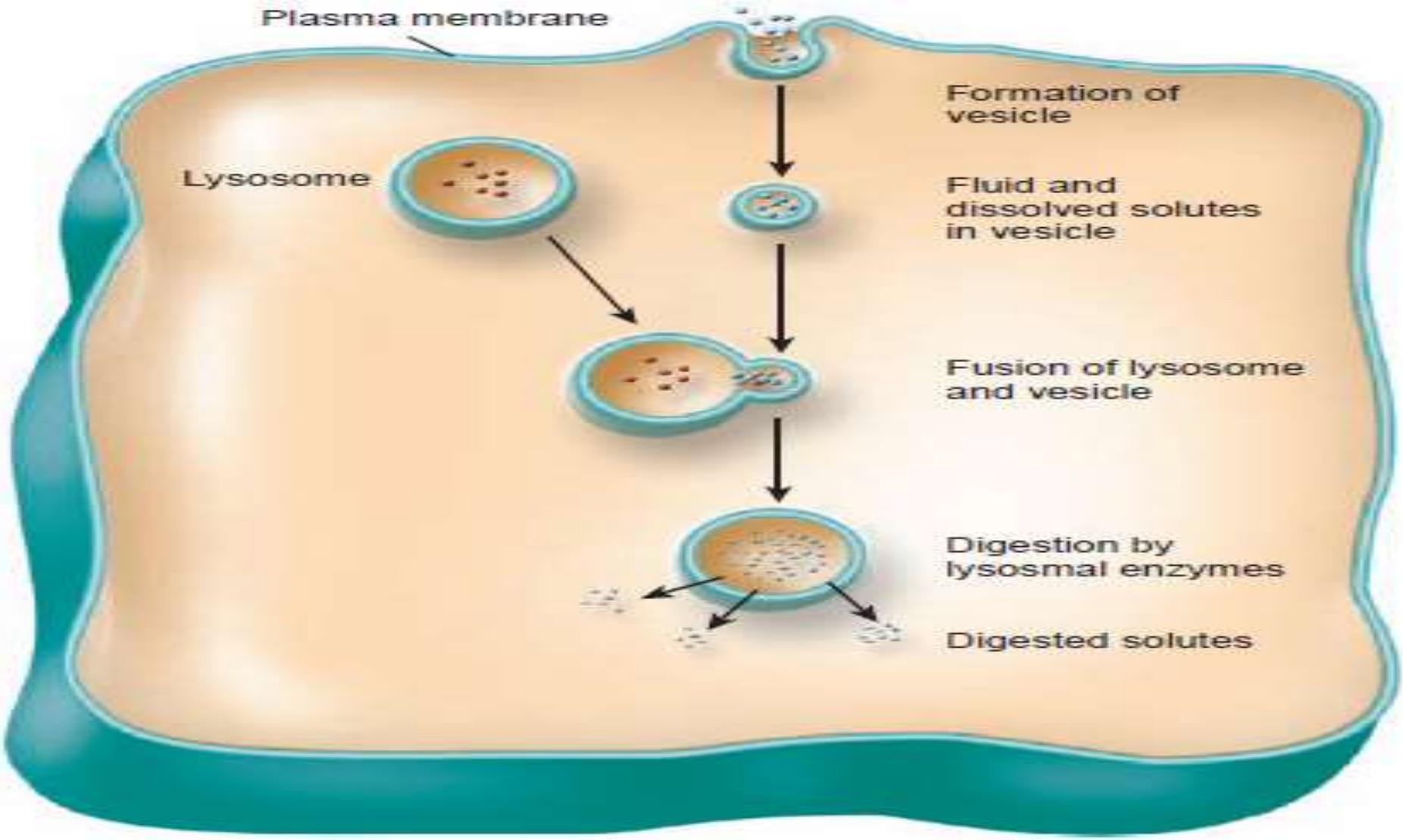
      ex- WBC – engulf bacteria
  - Pinocytosis- cell drinking

    Uptake of fluid / fluid contents into the cell.

    - occurs in all cells.

    ex- Uptake of proteins into cell

Endocytosis



Exocytosis

- ✓ Expulsing molecules out from the cell.
- ✓ Fate of molecule released by exocytosis may be……..
  - ✓ Peripheral proteins
  - ✓ Part of extracellular matrix
  - ✓ Released to extracellular medium

THANK YOU